EXTENDED SURVIVORSHIP CONSULTATION

REGIONAL CANCER INSTITUTE
Long-Term Follow-Up Clinic

COMPREHENSIVE SURVIVORSHIP CONSULTATION

Date of Consultation: March 19, 2024
Patient: Frank Phillips
MRN: AML049
DOB: 07/21/1958
Time in Remission: 28 months
Referring Physician: Dr. Michael Brown, Hematology

REASON FOR CONSULTATION: Comprehensive survivorship evaluation for a 65-year-old gentleman with core binding factor AML t(8;21) in sustained CR1 for 28 months, now transitioning from active surveillance to long-term survivorship care.

COMPREHENSIVE ONCOLOGIC HISTORY:

Mr. Phillips presented in November 2021 at age 63 with what he described as "the worst flu of my life" - fevers to 103°F, drenching night sweats, profound fatigue, and a 25-pound weight loss over 6 weeks. His wife, a retired nurse, insisted on medical evaluation when she noticed petechiae on his lower extremities.

Initial Evaluation (11/14/2021): Physical examination revealed a pale, ill-appearing gentleman with scattered petechiae and ecchymoses, mild splenomegaly, and no lymphadenopathy. Laboratory evaluation was striking for:

- WBC: 89,000/μL with 68% blasts
- Hemoglobin: 6.2 g/dL
- Platelets: 12,000/μL
- LDH: 2,341 U/L
- Uric acid: 12.3 mg/dL

Diagnostic Workup: Urgent bone marrow biopsy (11/15/2021) revealed:

- Morphology: 84% myeloblasts with prominent Auer rods

- Cytogenetics: t(8;21)(q22;q22.1) - RUNX1-RUNX1T1 fusion in 20/20 metaphases
- Molecular: RUNX1-RUNX1T1 fusion confirmed, KIT mutation D816V detected (associated with higher relapse risk in CBF-AML)
- Additional mutations: ASXL1, TET2 (likely age-related clonal hematopoiesis)
- Classification: Core binding factor AML with t(8;21), favorable risk despite KIT mutation

Emergency Management: Given impending leukostasis (WBC 89K) and tumor lysis syndrome, he required:

- Immediate hydroxyurea 3g PO
- Leukapheresis x2 sessions (reduced WBC to 34K)
- Rasburicase for hyperuricemia
- Aggressive hydration and electrolyte management

TREATMENT COURSE:

Induction with FLAG-Ida (November 21-26, 2021): Given his high disease burden and KIT mutation, the team opted for intensified induction:

- Fludarabine 30 mg/m² IV days 2-6
- Cytarabine 2 g/m² IV days 2-6 (4 hours after fludarabine)
- Idarubicin 10 mg/m² IV days 2-4
- G-CSF 300 mcg SC days 1-7

This aggressive approach was complicated by:

- Severe tumor lysis requiring continuous renal replacement therapy x48 hours
- Grade 4 mucositis necessitating TPN for 3 weeks
- Polymicrobial sepsis (E. coli and Candida glabrata) requiring 21 days of antimicrobials
- Prolonged pancytopenia lasting 42 days
- ICU admission days 8-14 for septic shock

Response Assessment:

- Day 14 marrow: Aplastic, no residual leukemia
- Day 35 marrow: CR achieved, 1% blasts, MRD negative by flow
- RUNX1-RUNX1T1 transcript: 2.5 log reduction from baseline

Consolidation Therapy (January-May 2022): Four cycles of high-dose cytarabine with close monitoring for KIT mutation:

Cycle 1: HiDAC 3 g/m² q12h days 1,3,5

- Complicated by severe conjunctivitis (cytarabine-related)
- Required steroid eye drops

Cycles 2-4: Dose reduced to 2 g/m² due to ocular toxicity

- Each cycle required brief hospitalization for neutropenic fever
- Cumulative platelet transfusions: 42 units
- Cumulative pRBC transfusions: 18 units

Post-Consolidation Assessment (June 2022):

- Bone marrow: Continued CR, MRD negative
- RUNX1-RUNX1T1 transcript: Undetectable (sensitivity 10^-5)
- KIT mutation: Still detectable at low level (VAF 0.8%)

LONG-TERM FOLLOW-UP COURSE:

Months 6-12 (June-December 2022):

- Monthly CBC and RUNX1-RUNX1T1 PCR monitoring
- Bone marrow biopsies at months 9 and 12: Continued CR, MRD negative
- Gradual recovery of counts to normal range
- Returned to work part-time as financial advisor

Months 12-24 (December 2022-December 2023):

- Monitoring decreased to every 2-3 months
- Serial PCR remained negative
- KIT mutation became undetectable at month 18

- Full return to work and normal activities
- Completed cardiac rehabilitation program (preventive)

Current Status at 28 Months:

- Remains in complete molecular remission
- No evidence of minimal residual disease
- Excellent performance status (ECOG 0)
- Has resumed all activities including international travel

COMPREHENSIVE SURVIVORSHIP ASSESSMENT:

Current Symptoms Review: Mr. Phillips reports feeling "95% back to normal" with the following residual issues:

- Mild peripheral neuropathy in fingertips (improved with gabapentin 300mg TID)
- Occasional "chemo brain" - word-finding difficulties, improved with cognitive exercises
- Mild exertional dyspnea (likely deconditioning vs early cardiotoxicity)
- Decreased libido (multifactorial - psychological, physiological, relationship stress)

Physical Examination:

- Vital signs: BP 128/78, HR 72, Weight 178 lbs (recovered from nadir of 152 lbs)
- General: Well-appearing gentleman, appears younger than stated age
- HEENT: Mild bilateral cataracts (steroid-related), no active conjunctivitis
- Cardiovascular: Regular rhythm, no murmurs, trace pedal edema
- Pulmonary: Clear to auscultation
- Abdomen: Soft, no organomegaly
- Neurologic: Decreased sensation to light touch in distal fingers/toes
- Skin: Multiple areas of hyperpigmentation (post-inflammatory)

Laboratory Studies (03/15/2024):

- CBC: WBC 5,800 (normal differential), Hgb 14.2 g/dL, Platelets 198,000
- Chemistry: Creatinine 1.1 mg/dL (baseline 0.9), LFTs normal
- Thyroid function: Normal (post-chemotherapy screening)
- Testosterone: 285 ng/dL (low normal)
- Lipid panel: Total cholesterol 218, LDL 142, HDL 38, Triglycerides 189
- HbA1c: 5.9%
- RUNX1-RUNX1T1 transcript: Undetectable
- Bone marrow (03/01/2024): Normocellular, no evidence of AML, normal cytogenetics